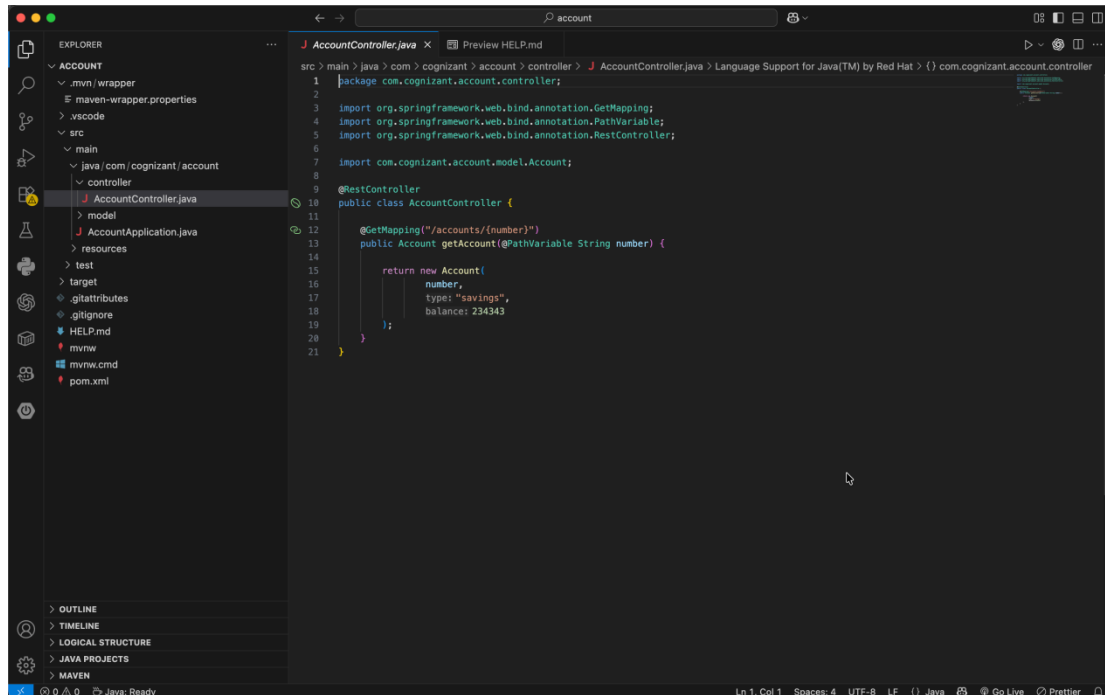


DIGITAL NURTURE 5.0 DEEPSKILLING

WEEK 4 – HANDS-ON

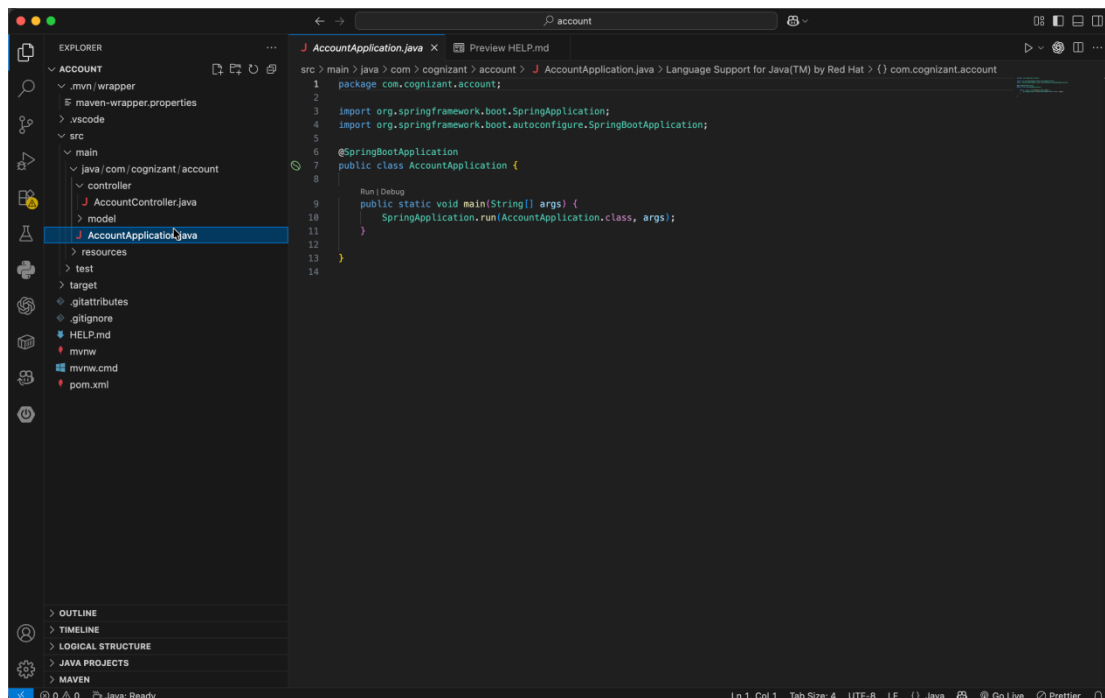
Micro services with Spring Boot 3 and Spring Cloud

ACCOUNT



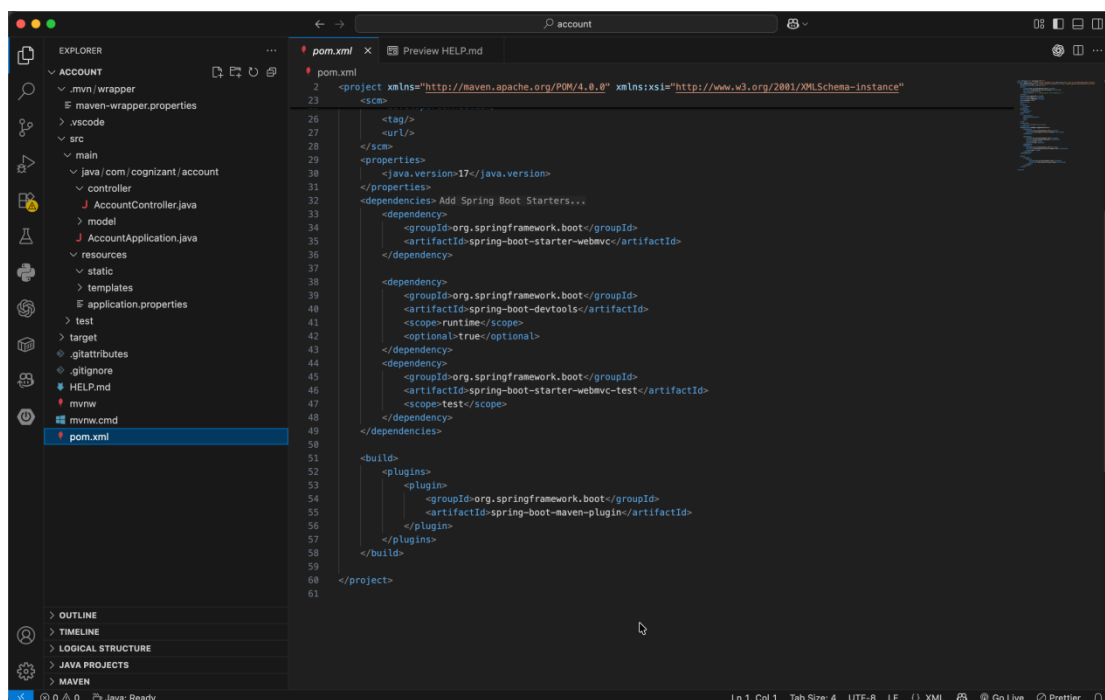
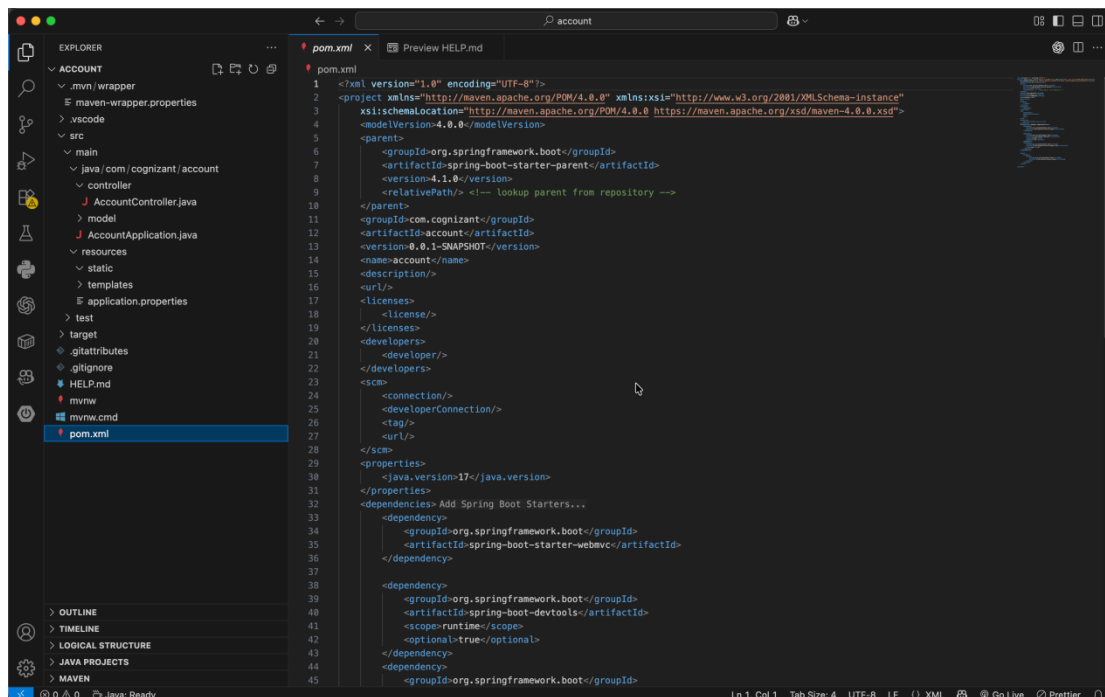
This screenshot shows the Visual Studio Code editor with the `AccountController.java` file open. The Explorer sidebar on the left shows the project structure, with the `controller` folder selected. The main editor displays the following Java code:

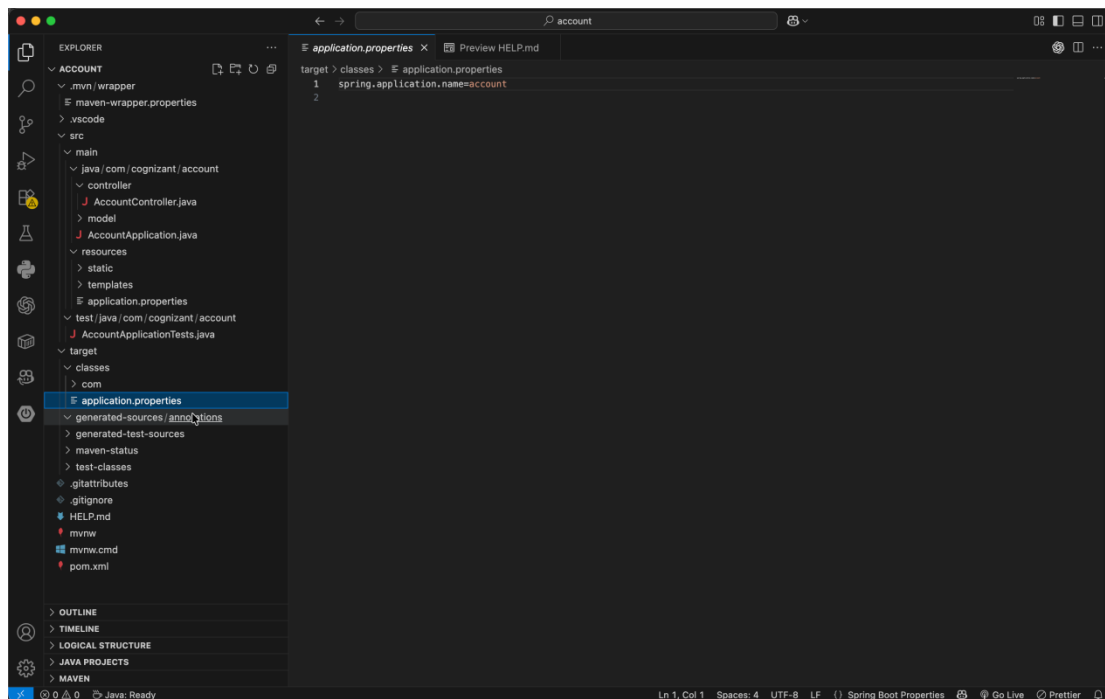
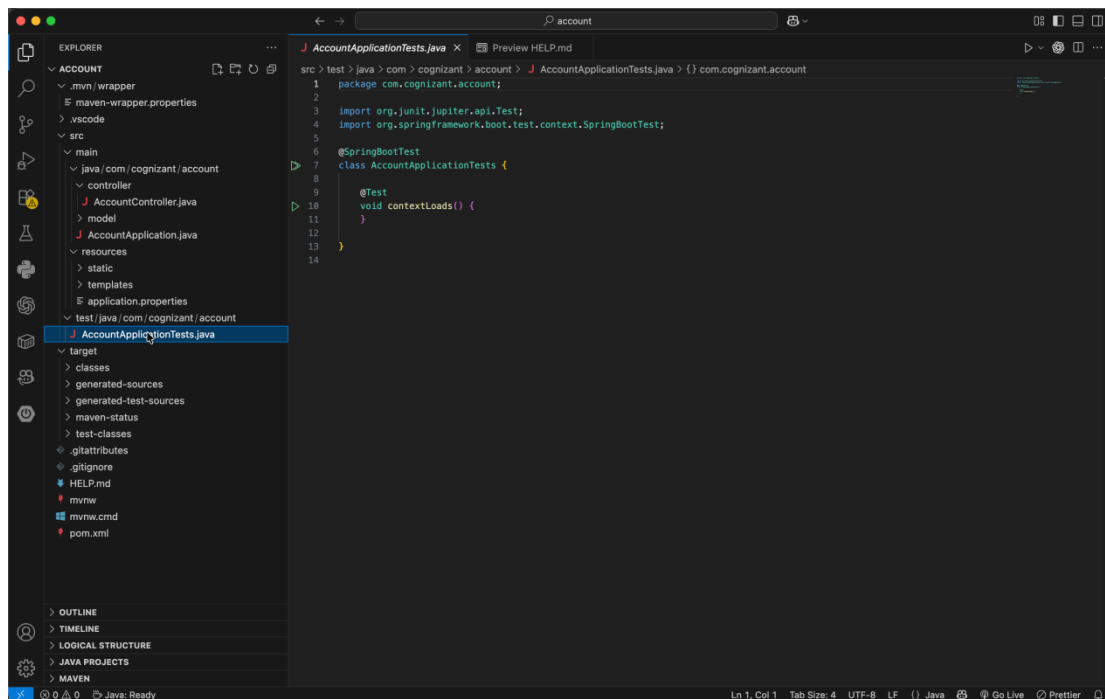
```
src > main > java > com > cognizant > account > controller > J AccountController.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.account.controller
1 package com.cognizant.account.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.account.model.Account;
8
9 @RestController
10 public class AccountController {
11
12     @GetMapping("/accounts/{number}")
13     public Account getAccount(@PathVariable String number) {
14
15         return new Account(
16             number,
17             type: "savings",
18             balance: 234343
19         );
20     }
21 }
```



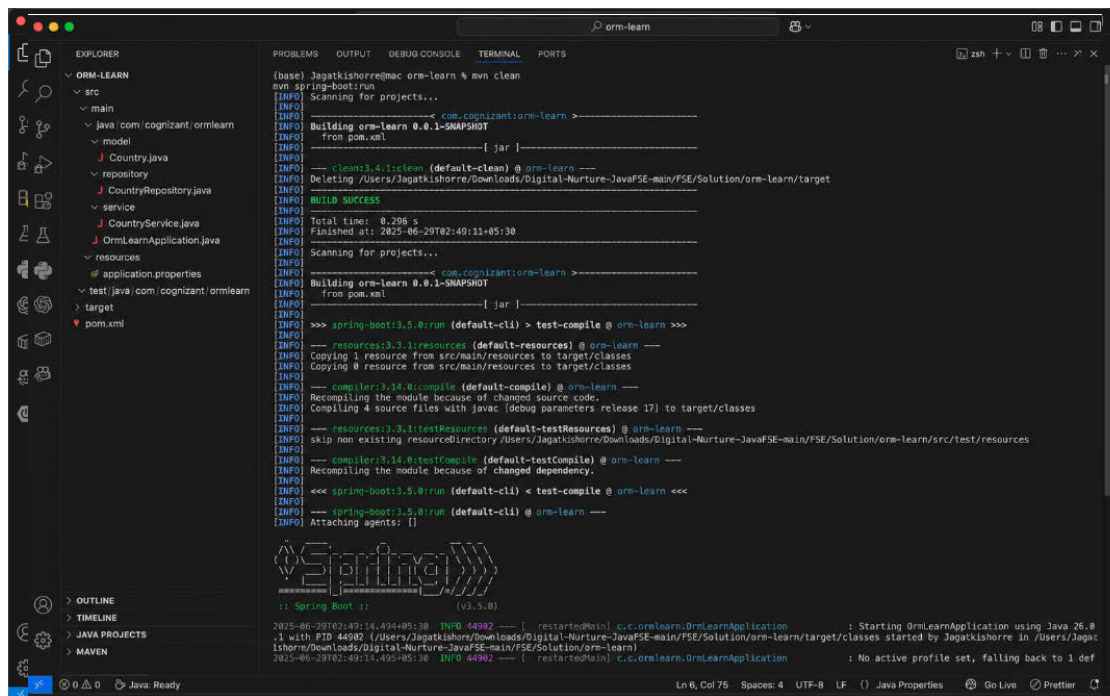
This screenshot shows the Visual Studio Code editor with the `AccountApplication.java` file open. The Explorer sidebar on the left shows the project structure, with the `model` folder selected. The main editor displays the following Java code:

```
src > main > java > com > cognizant > account > J AccountApplication.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.account
1 package com.cognizant.account;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6
7 @SpringBootApplication
8 public class AccountApplication {
9
10     Run | Debug
11     public static void main(String[] args) {
12         SpringApplication.run(AccountApplication.class, args);
13     }
14 }
```





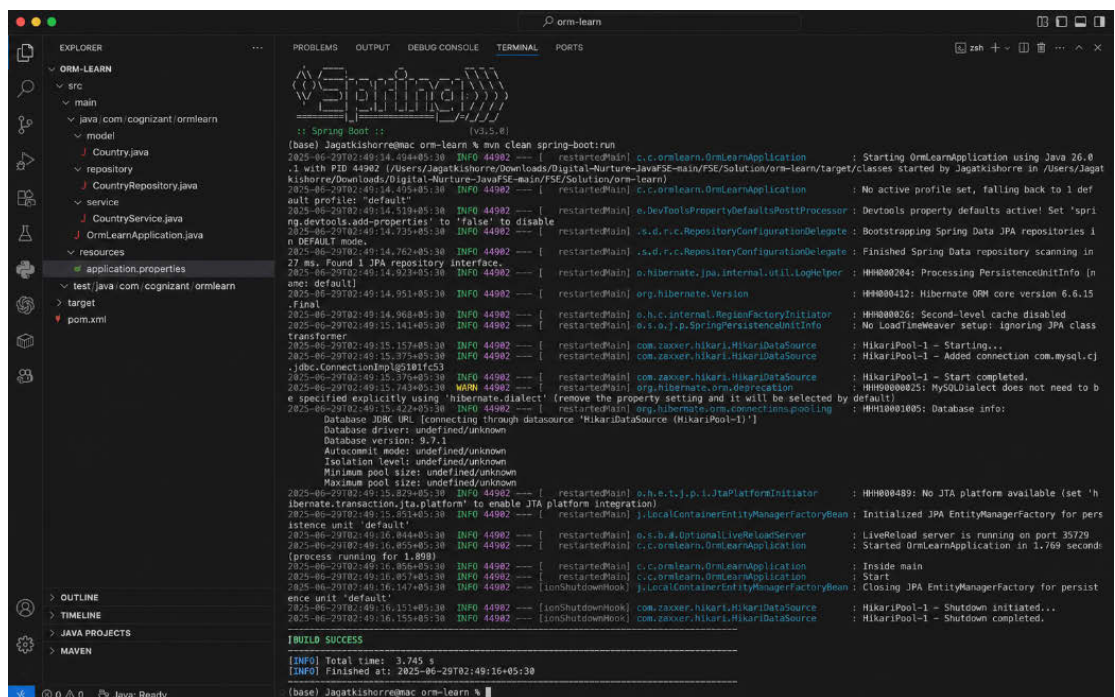
OUTPUT



The screenshot shows an IDE window titled 'orm-learn'. The left sidebar contains an 'EXPLORER' view with a project structure for 'ORM-LEARN' including 'src', 'main', 'model', 'Country.java', 'CountryRepository.java', 'service', 'CountryService.java', 'OrmLearnApplication.java', 'resources', 'application.properties', 'test/java/com/cognizant/ormlearn', 'target', and 'pom.xml'. The main area is divided into 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', and 'TERMINAL' tabs. The 'TERMINAL' tab is active, displaying the build output of a Spring Boot application. The output shows the following steps:

- Scanning for projects...
- Building orm-learn 0.0.1-SNAPSHOT from pom.xml
- Deleting /Users/Jagatkishor/downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn/target
- BUILD SUCCESS
- Total time: 0.286 s
- Finished at: 2025-06-29T02:49:11+05:30
- Scanning for projects...
- Building orm-learn 0.0.1-SNAPSHOT from pom.xml
- spring-boot:3.5.0:run (default-cli) > test-compile @ orm-learn >>>
- resources:3.3.1:resources (default-resources) @ orm-learn ---
- Copying 1 resource from src/main/resources to target/classes
- Copying 0 resource from src/main/resources to target/classes
- compiler:3.14.0:compile (default-compile) @ orm-learn ---
- Recompiling the module because of changed source code.
- Compiling 4 source files with javac (Debug parameters: release 17) to target/classes
- resources:3.3.1:testResources (default-testResources) @ orm-learn ---
- skip non existing resourceDirectory /Users/Jagatkishor/downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn/src/test/resources
- compiler:3.14.0:testCompile (default-testCompile) @ orm-learn ---
- Recompiling the module because of changed dependency.
- spring-boot:3.5.0:run (default-cli) < test-compile @ orm-learn <<<
- spring-boot:3.5.0:run (default-cli) @ orm-learn ---
- Attaching agents: []

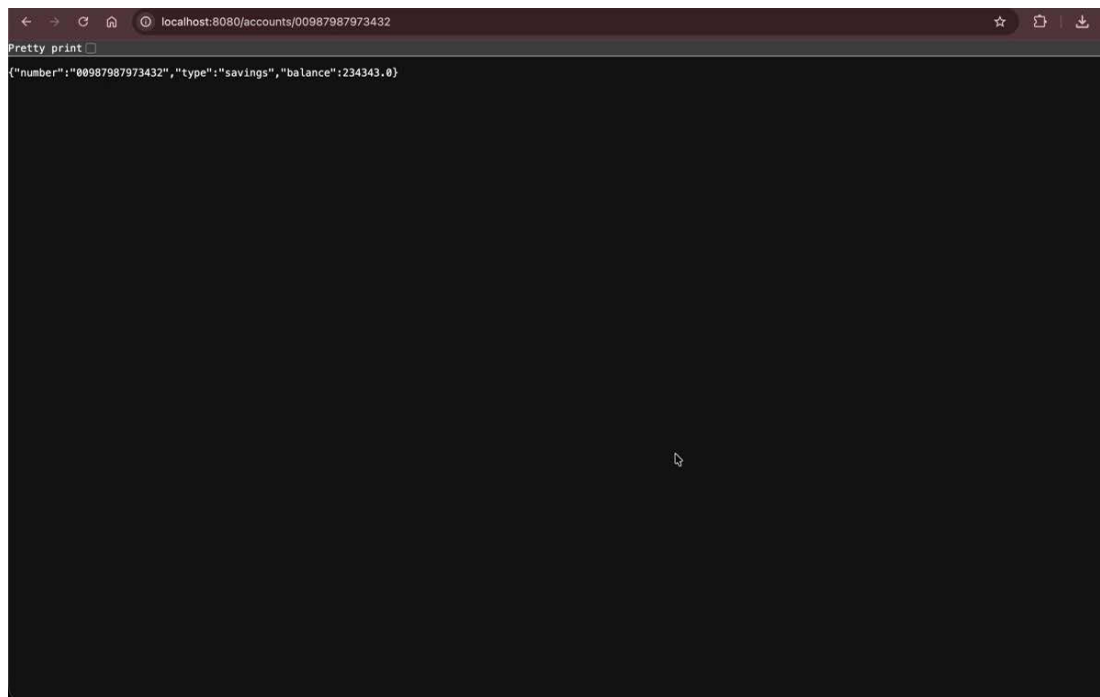
The output also shows the Spring Boot logo and the version (v3.5.0). The bottom status bar indicates 'Ln 6, Col 75 Spaces: 4 UTF-8 LF Java Properties Go Live Prettier'.



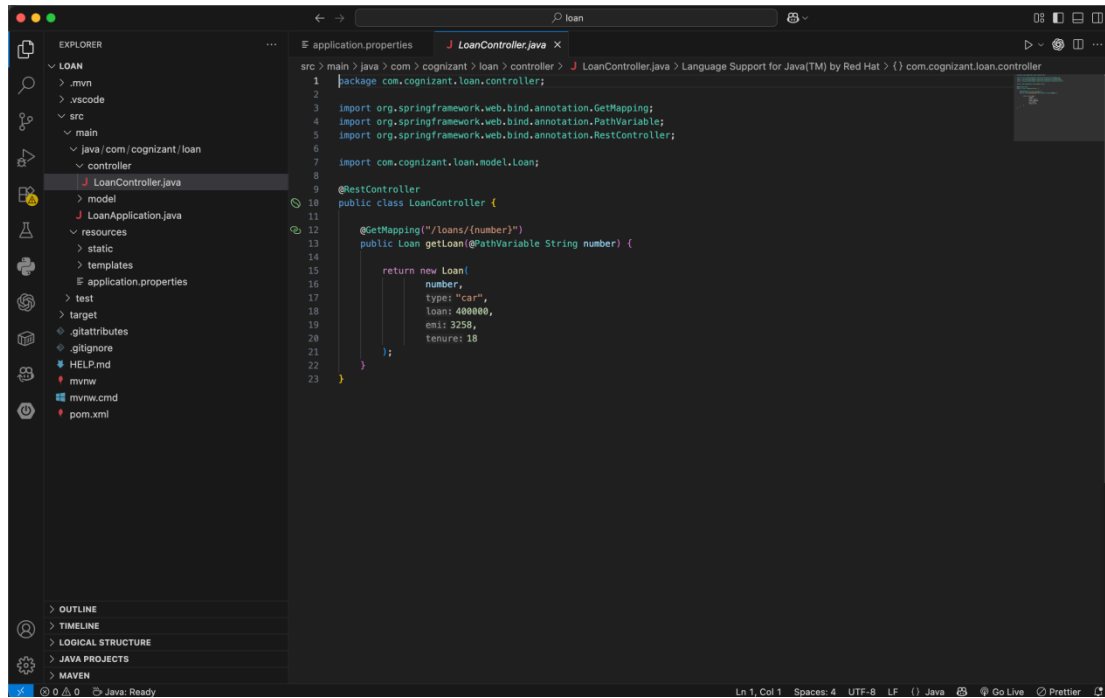
The screenshot shows the same IDE window as above, but now displaying the runtime output of the Spring Boot application. The 'TERMINAL' tab is active, showing the following steps:

- Spring Boot v3.5.0
- (base) Jagatkishor@mac-orm-learn % mvn clean spring-boot:run
- 2025-06-29T02:49:14.494+05:30 INFO 44982 [restartedMain] c.c.ormlearn.OrmLearnApplication : Starting OrmLearnApplication using Java 26.0
- .1 with PID 44982 (/Users/Jagatkishor/downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn/target/classes started by Jagatkishor in /Users/Jagatkishor/downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn)
- 2025-06-29T02:49:14.505+05:30 INFO 44982 [restartedMain] c.c.ormlearn.OrmLearnApplication : No active profile set, falling back to 1 def
- ault profile: "default"
- 2025-06-29T02:49:14.519+05:30 INFO 44982 [restartedMain] o.devtools.properties.DefaultPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
- 2025-06-29T02:49:14.735+05:30 INFO 44982 [restartedMain] s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in
- n DEFAULT mode.
- 2025-06-29T02:49:14.762+05:30 INFO 44982 [restartedMain] s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in
- 27 ms. Found 1 JPA repository interface.
- 2025-06-29T02:49:14.933+05:30 INFO 44982 [restartedMain] o.hibernate.jpa.internal.util.LogHelper : HH0000204: Processing PersistenceUnitInfo [n
- ame: default]
- 2025-06-29T02:49:14.951+05:30 INFO 44982 [restartedMain] org.hibernate.Version : HH0000412: Hibernate ORM core version 6.6.15
- .Final
- 2025-06-29T02:49:14.968+05:30 INFO 44982 [restartedMain] o.h.c.i.internal.RegionFactoryInitiator : HH0000026: Second-level cache disabled
- 2025-06-29T02:49:15.141+05:30 INFO 44982 [restartedMain] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class
- transformer
- 2025-06-29T02:49:15.157+05:30 INFO 44982 [restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
- 2025-06-29T02:49:15.273+05:30 INFO 44982 [restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Added connection com.mysql.cj
- ydbc.ConnectionImpl@181fc53
- 2025-06-29T02:49:15.376+05:30 INFO 44982 [restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
- 2025-06-29T02:49:15.743+05:30 WARN 44982 [restartedMain] org.hibernate.orm.deprecation : HH0000025: MySQLDialect does not need to b
- e specified explicitly using 'hibernate.dialect' (remove the property setting and it will be selected by default)
- 2025-06-29T02:49:15.822+05:30 INFO 44982 [restartedMain] org.hibernate.orm.connections.pooling : HH0000005: Database info:
- Database driver: undefined/unknown
- Database version: 9.7.1
- AutoCommit mode: undefined/unknown
- Isolation level: undefined/unknown
- Minimum pool size: undefined/unknown
- Maximum pool size: undefined/unknown
- 2025-06-29T02:49:15.879+05:30 INFO 44982 [restartedMain] o.h.e.t.j.p.l.JtaPlatformInitiator : HH0000489: No JTA platform available (set 'h
- ibernate.transaction.jta.platform' to enable JTA platform integration)
- 2025-06-29T02:49:15.885+05:30 INFO 44982 [restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for pers
- istence unit 'default'
- 2025-06-29T02:49:16.044+05:30 INFO 44982 [restartedMain] o.s.b.o.OptionalLiveReloadServer : LiveReload server is running on port 35729
- 2025-06-29T02:49:16.053+05:30 INFO 44982 [restartedMain] c.c.ormlearn.OrmLearnApplication : Started OrmLearnApplication in 1.769 seconds.
- 2025-06-29T02:49:16.056+05:30 INFO 44982 [restartedMain] c.c.ormlearn.OrmLearnApplication : Inside main
- 2025-06-29T02:49:16.074+05:30 INFO 44982 [restartedMain] j.LocalContainerEntityManagerFactoryBean : Started
- 2025-06-29T02:49:16.147+05:30 INFO 44982 [restartedMain] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persist
- ence unit 'default'
- 2025-06-29T02:49:16.151+05:30 INFO 44982 [restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
- 2025-06-29T02:49:16.155+05:30 INFO 44982 [restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.

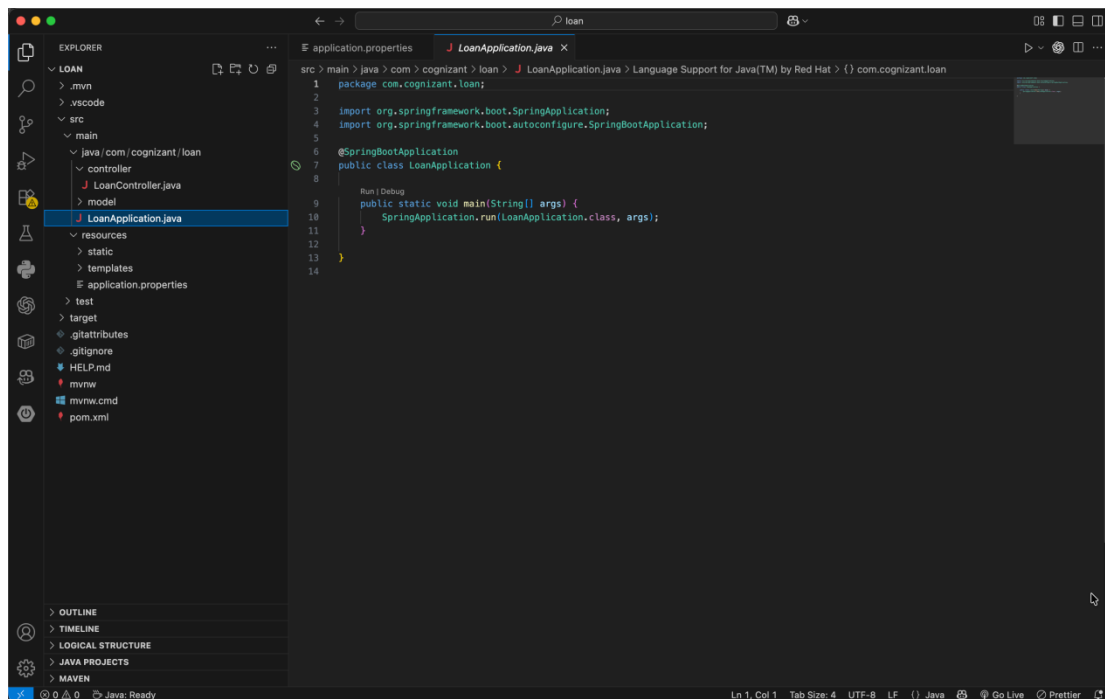
The output also shows the Spring Boot logo and the version (v3.5.0). The bottom status bar indicates 'Ln 6, Col 75 Spaces: 4 UTF-8 LF Java Properties Go Live Prettier'.



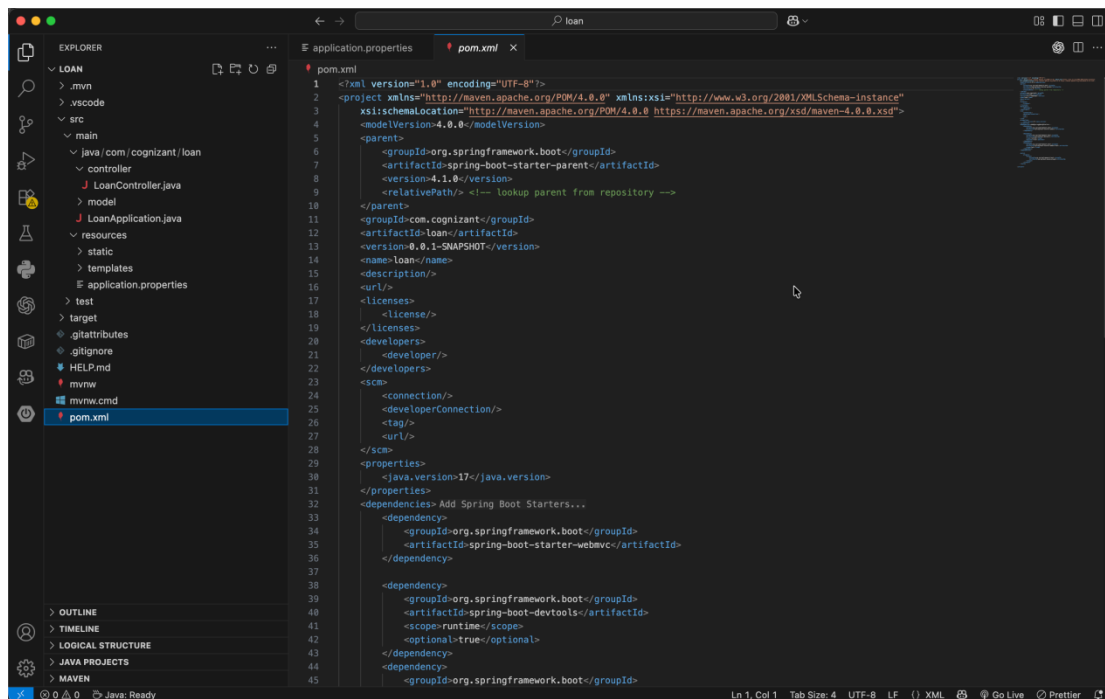
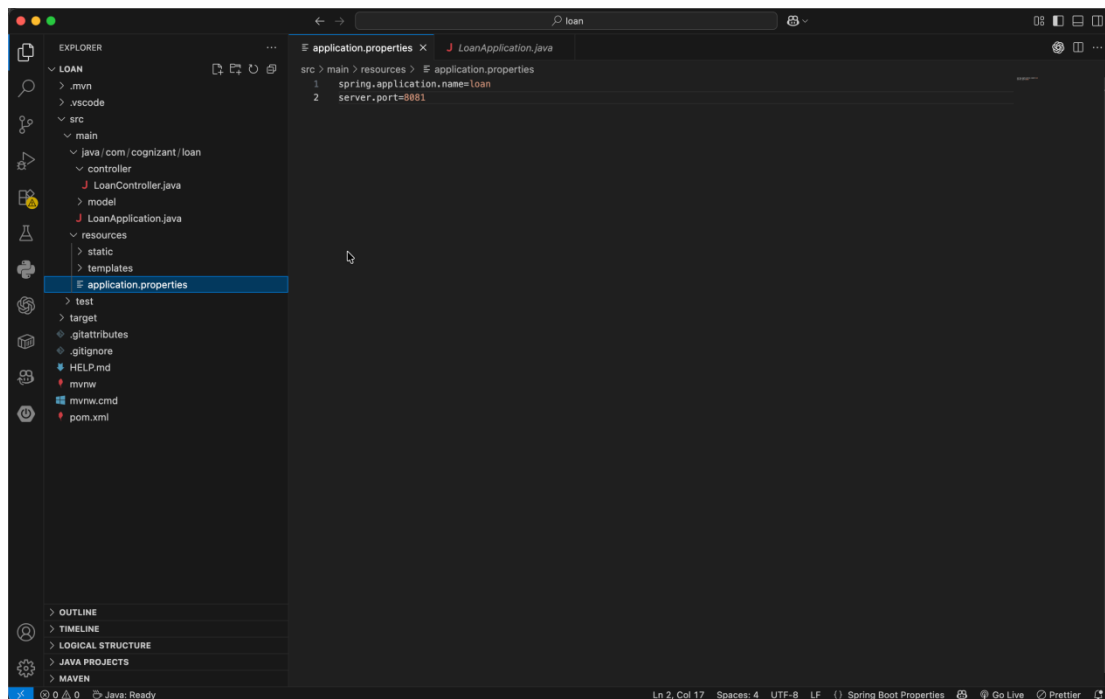
LOAN

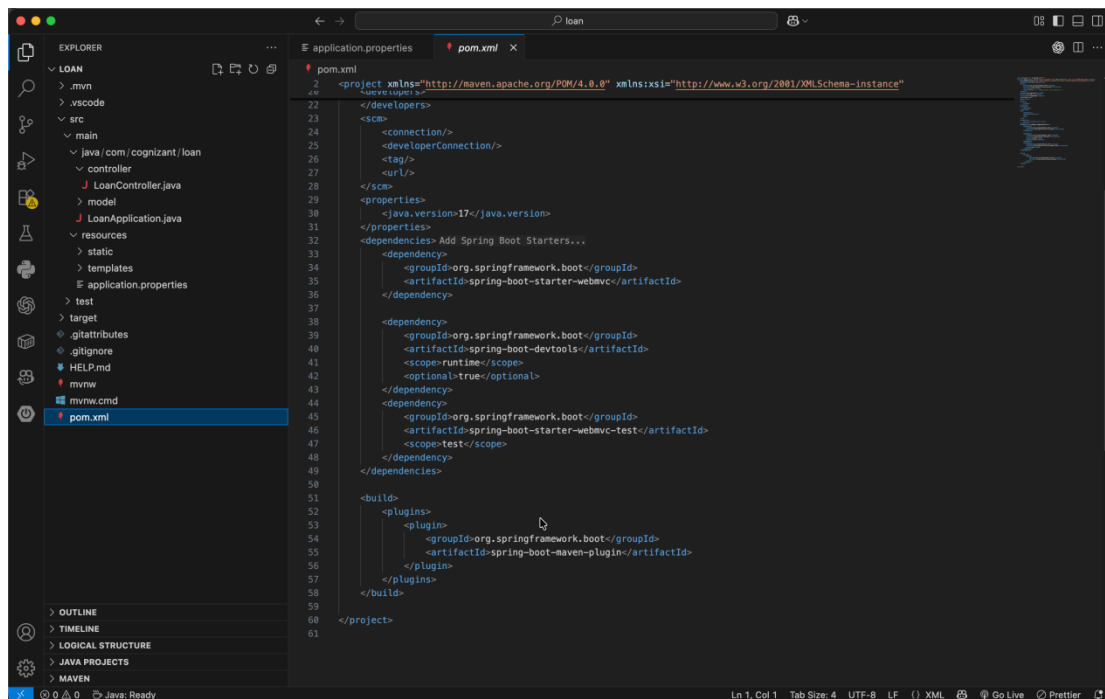


```
src > main > java > com > cognizant > loan > controller > J LoanController.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.loan.controller
1 package com.cognizant.loan.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.loan.model.Loan;
8
9 @RestController
10 public class LoanController {
11
12     @GetMapping("/loans/{number}")
13     public Loan getLoan(@PathVariable String number) {
14
15         return new Loan(
16             number,
17             type: "car",
18             loan: 400000,
19             emi: 3258,
20             tenure: 18
21         );
22     }
23 }
```

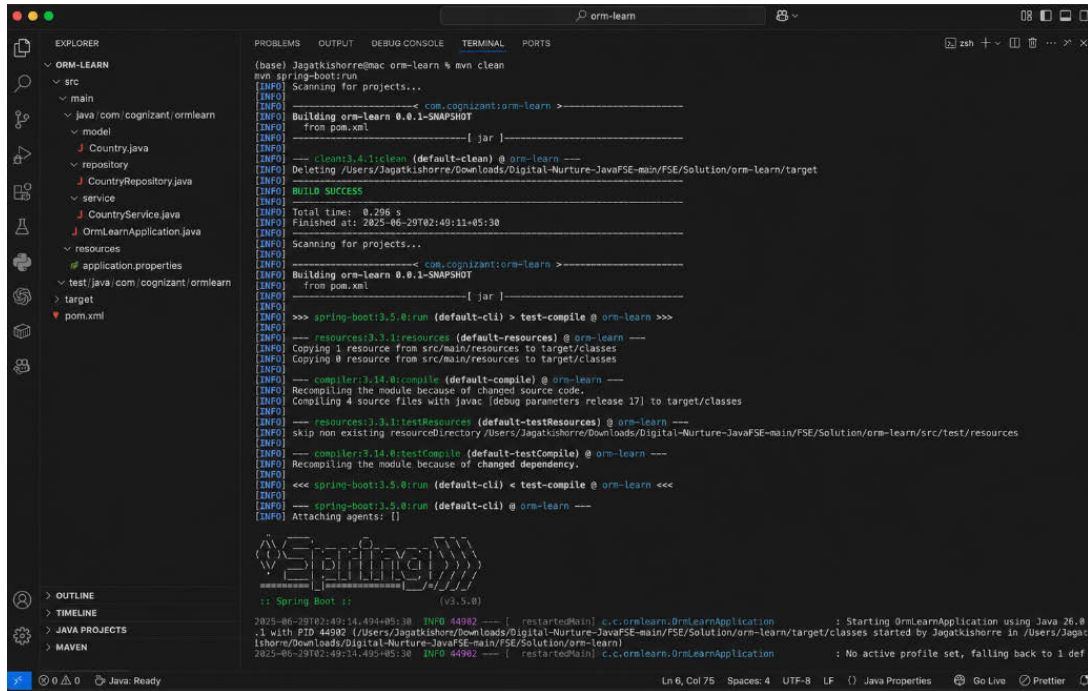


```
src > main > java > com > cognizant > loan > J LoanApplication.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.loan
1 package com.cognizant.loan;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class LoanApplication {
8
9     Run | Debug
10     public static void main(String[] args) {
11         SpringApplication.run(LoanApplication.class, args);
12     }
13
14 }
```





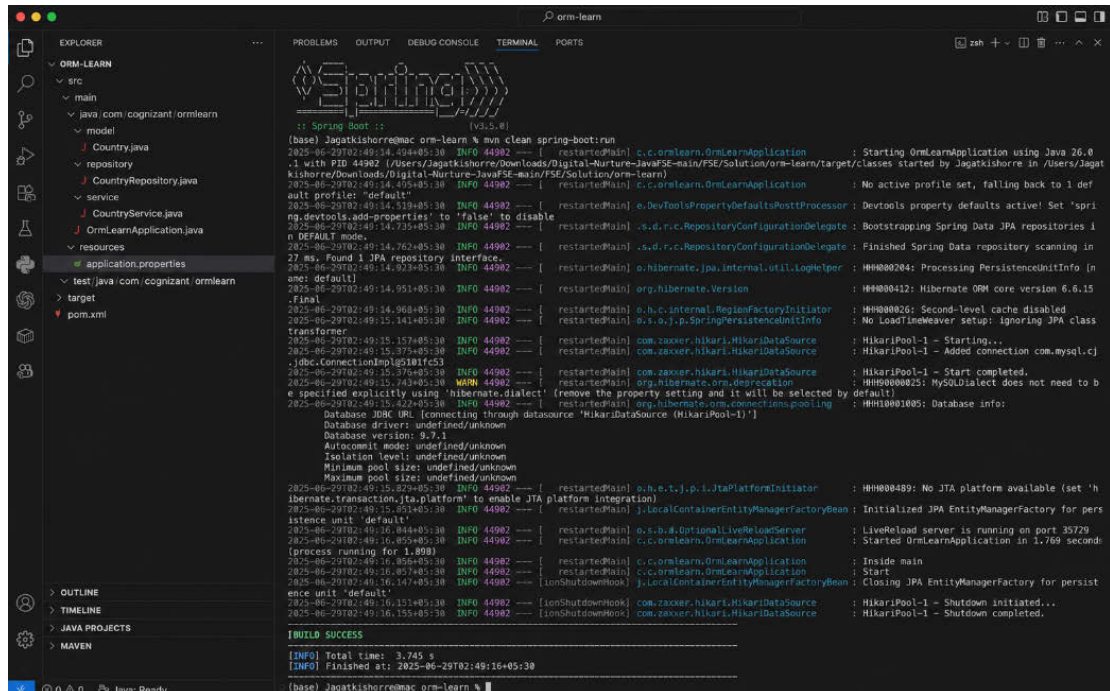
OUTPUT



```
(base) Jagatkishor@mac: orm-learn % mvn clean
mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] < com.cognizant:orm-learn >
[INFO] from pom.xml
[INFO]
[INFO] --- clean:3.4.1:clean (default-clean) @ orm-learn ---
[INFO] Deleting /Users/Jagatkishor/Downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn/target
[INFO] BUILD SUCCESS
[INFO] Total time: 0.296 s
[INFO] Finished at: 2025-06-29T02:49:11+05:30
[INFO] Scanning for projects...
[INFO]
[INFO] < com.cognizant:orm-learn >
[INFO] from pom.xml
[INFO]
[INFO] --- spring-boot:3.5.0:run (default-cli) @ orm-learn ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ orm-learn ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.14.0:compile (default-compile) @ orm-learn ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 4 source files with javac [debug parameters release 17] to target/classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ orm-learn ---
[INFO] skip non existing resourceDirectory /Users/Jagatkishor/Downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn/src/test/resources
[INFO]
[INFO] --- compiler:3.14.0:testCompile (default-testCompile) @ orm-learn ---
[INFO] Recompiling the module because of changed dependency.
[INFO]
[INFO] << spring-boot:3.5.0:run (default-cli) < test-compile @ orm-learn <<
[INFO]
[INFO] --- spring-boot:3.5.0:run (default-cli) @ orm-learn ---
[INFO] Attaching agents: []

Spring Boot (v3.5.0)

2025-06-29T02:49:14.494+05:30 INFO 44902 [ restartedMain] c.c.ormlearn.OrmLearnApplication : Starting OrmLearnApplication using Java 26.0
.1 with PID 44902 (/Users/Jagatkishor/Downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn/target/classes started by Jagatkishor in /Users/Jagatkishor/Downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn)
2025-06-29T02:49:14.495+05:30 INFO 44902 [ restartedMain] c.c.ormlearn.OrmLearnApplication : No active profile set, falling back to 1 def
```



```

Spring Boot (v3.5.0)

2025-06-29T02:49:14.494+05:30 INFO 44902 [ restartedMain] c.c.ormlearn.OrmLearnApplication : Starting OrmLearnApplication using Java 26.0
.1 with PID 44902 (/Users/Jagatkishor/Downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn/target/classes started by Jagatkishor in /Users/Jagatkishor/Downloads/Digital-Nurture-JavaFSE-main/FSE/Solution/orm-learn)
2025-06-29T02:49:14.495+05:30 INFO 44902 [ restartedMain] c.c.ormlearn.OrmLearnApplication : No active profile set, falling back to 1 def
ault profiles: "default"
2025-06-29T02:49:14.519+05:30 INFO 44902 [ restartedMain] o.e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2025-06-29T02:49:14.725+05:30 INFO 44902 [ restartedMain] s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories i
n DEFAULT mode.
2025-06-29T02:49:14.762+05:30 INFO 44902 [ restartedMain] s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in
27 ms. Found 1 JPA repository interface.
2025-06-29T02:49:14.923+05:30 INFO 44902 [ restartedMain] o.hiber.nate.jpa.internal.util.LogHelper : HH0000204: Processing PersistenceUnitInfo [n
ame: default]
2025-06-29T02:49:14.951+05:30 INFO 44902 [ restartedMain] org.hibernate.Version : HH0000412: Hibernate ORM core version 6.6.15
.Final
2025-06-29T02:49:14.968+05:30 INFO 44902 [ restartedMain] o.h.c.internal.RegionFactoryInitiator : HH0000026: Second-level cache disabled
2025-06-29T02:49:15.141+05:30 INFO 44902 [ restartedMain] o.s.a.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class
transformer
2025-06-29T02:49:15.157+05:30 INFO 44902 [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2025-06-29T02:49:15.375+05:30 INFO 44902 [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Added connection com.mysql.cj
.jdbc.ConnectionImpl@181fc53
2025-06-29T02:49:15.376+05:30 INFO 44902 [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2025-06-29T02:49:15.743+05:30 WARN 44902 [ restartedMain] org.hibernate.orm.deprecation : HH00000025: MySQLDialect does not need to b
e specified explicitly using 'hibernate.dialect' (remove the property setting and it will be selected by default)
2025-06-29T02:49:15.823+05:30 INFO 44902 [ restartedMain] org.hibernate.orm.connections.pooling : HH00001005: Database info:
Database JDBC URL [connecting through datasource 'HikariDataSource (HikariPool-1)']
Database driver: undefined/unknown
Database version: 9.7.1
Autocommit mode: undefined/unknown
Isolation level: undefined/unknown
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2025-06-29T02:49:15.823+05:30 INFO 44902 [ restartedMain] o.s.e.t.j.p.i.JtaPlatformInitiator : HH0000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2025-06-29T02:49:15.854+05:30 INFO 44902 [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for pers
istence unit 'default'
2025-06-29T02:49:16.864+05:30 INFO 44902 [ restartedMain] o.s.b.o.OptionalLiveReloadServer : LiveReload server is running on port 35729
2025-06-29T02:49:16.865+05:30 INFO 44902 [ restartedMain] c.c.ormlearn.OrmLearnApplication : Started OrmLearnApplication in 1.769 seconds
(process running for 1.899)
2025-06-29T02:49:16.864+05:30 INFO 44902 [ restartedMain] c.c.ormlearn.OrmLearnApplication : Inside main
2025-06-29T02:49:16.857+05:30 INFO 44902 [ restartedMain] c.c.ormlearn.OrmLearnApplication : Start
2025-06-29T02:49:16.147+05:30 INFO 44902 [ shutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persist
ence unit 'default'
2025-06-29T02:49:16.151+05:30 INFO 44902 [ shutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2025-06-29T02:49:16.155+05:30 INFO 44902 [ shutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.

BUILD SUCCESS
[INFO] Total time: 3.745 s
[INFO] Finished at: 2025-06-29T02:49:16+05:30
(base) Jagatkishor@mac: orm-learn %
```

```
(base) Jagatkishorremac orm-learn % mvn clean spring-boot:run
2025-06-29T02:49:14.494+05:30 INFO 44982 --- [ restartedMain] c.c.ormlearn.OrmLearnApplication : Starting OrmLearnApplication using Java 26.0
2025-06-29T02:49:14.535+05:30 INFO 44982 --- [ restartedMain] o.s.b.f.w.WebFluxContextLoader : No active profile set, falling back to 1 default profile: 'default'
2025-06-29T02:49:14.519+05:30 INFO 44982 --- [ restartedMain] o.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2025-06-29T02:49:14.535+05:30 INFO 44982 --- [ restartedMain] o.s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2025-06-29T02:49:14.763+05:30 INFO 44982 --- [ restartedMain] o.s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 27 ms. Found 1 JPA repository interface.
2025-06-29T02:49:14.931+05:30 INFO 44982 --- [ restartedMain] o.hibernate.jpa.internal.util.LogHelper : HHH0000204: Processing PersistenceUnitInfo [name: default]
2025-06-29T02:49:14.951+05:30 INFO 44982 --- [ restartedMain] org.hibernate.Version : HHH0000412: Hibernate ORM core version 6.6.15
2025-06-29T02:49:14.968+05:30 INFO 44982 --- [ restartedMain] o.h.c.internal.RegionFactoryInitiator : HHH0000026: Second-level cache disabled
2025-06-29T02:49:15.141+05:30 INFO 44982 --- [ restartedMain] o.s.a.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup; ignoring JPA class transformer
2025-06-29T02:49:15.157+05:30 INFO 44982 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2025-06-29T02:49:15.373+05:30 INFO 44982 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@181f1653
2025-06-29T02:49:15.376+05:30 INFO 44982 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2025-06-29T02:49:15.743+05:30 WARN 44982 --- [ restartedMain] org.hibernate.orm.deprecation : HHH00000825: MySQLDialect does not need to be specified explicitly using 'hibernate.dialect' (remove the property setting and it will be selected by default)
2025-06-29T02:49:15.822+05:30 INFO 44982 --- [ restartedMain] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [connecting through datasource 'HikariDataSource (HikariPool-1)']
Database version: 9.7.1
Autocommit mode: undefined/unknown
Isolation level: undefined/unknown
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2025-06-29T02:49:15.829+05:30 INFO 44982 --- [ restartedMain] o.h.e.t.j.p.l.JtaPlatformInitiator : HHH0000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2025-06-29T02:49:15.853+05:30 INFO 44982 --- [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2025-06-29T02:49:16.044+05:30 INFO 44982 --- [ restartedMain] o.s.b.a.OptionalLiveReloadServer : LiveReload server is running on port 30729
2025-06-29T02:49:16.405+05:30 INFO 44982 --- [ restartedMain] c.c.ormlearn.OrmLearnApplication : Started OrmLearnApplication in 1.769 seconds (process running for 1.898)
2025-06-29T02:49:16.808+05:30 INFO 44982 --- [ restartedMain] c.c.ormlearn.OrmLearnApplication : Inside main
2025-06-29T02:49:16.857+05:30 INFO 44982 --- [ restartedMain] c.c.ormlearn.OrmLearnApplication : Start
2025-06-29T02:49:16.147+05:30 INFO 44982 --- [onShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'default'
2025-06-29T02:49:16.151+05:30 INFO 44982 --- [onShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2025-06-29T02:49:16.155+05:30 INFO 44982 --- [onShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.

BUILD SUCCESS

[INFO] Total time: 3.745 s
[INFO] Finished at: 2025-06-29T02:49:16+05:30
(base) Jagatkishorremac orm-learn %
```

```
localhost:8081/loans/H00987987972342
Pretty print
{"number":"H00987987972342","type":"car","loan":400000.0,"emi":3258,"tenure":18}
```

